FORS 491-02 74644 (Autumn 2025)

Please refer to the syllabus on Canvas for the most up-to-date version of this document. Some minor changes may be implemented over the course of the semester.

Course Syllabus

Instructor

Jonathan Lamont Batchelor *(they/them)*.

Jonathan.Batchelor@UMontana.edu, However, Please use Canvas for all communication about the course.

- Office hours: Tuesday 2pm to 4pm **CHCB 449**

Lecture / Lab

Lab: Monday 13:00 - 16:50, Stone 106

- We will meet in Stone 106 for all lab sessions; however, during the early part of the semester, most lab sessions will involve traveling to Lubrecht Forest.

Lecture: Thursday 14:00 - 14:50, Stone 106

Description

This applied course immerses students in the tools and techniques at the core of precision forestry and modern geospatial analysis. While we will meet in the computer lab for scheduled sessions, many early labs will be field trips—primarily to forest and wetland sites—for real-world data collection. Students will design and implement digital field surveys, collect high-accuracy GNSS data with differential correction, and acquire measurements using terrestrial laser scanning (TLS) and drone-based photogrammetry.

Airborne lidar (ALS) datasets, along with NAIP and Sentinel imagery, will be used to model forest metrics across landscapes.

Lab work emphasizes the integration of field and remotely sensed data using both ESRI tools—such as Survey123 and Dashboards—and open-source software, including R packages for lidar and machine learning. Students will process, analyze, and visualize spatial data, culminating in interactive dashboards, story maps, and final project presentations. While forests and wetlands are the course focus, the technical skills learned are broadly transferable across environmental and natural resource contexts.

Course Objectives

By the end of this course, students will:

1. Collect and differentially correct GNSS data, and compare accuracy across devices.
2. Operate TLS scanners, process and align point clouds using **CloudCompare**.
3. Plan and conduct UAV surveys, process imagery with photogrammetry workflows, and generate 3D products.
4. Acquire and integrate **ALS, NAIP, and Sentinel** datasets for forest and wetland analysis.
5. Use **Survey123 and Dashboards** to design digital surveys and communicate field results.
6. Apply **R** to process lidar data, derive canopy and tree-level metrics, and perform regression and machine learning analyses.

Learning Outcomes

Upon successful completion of the course, students will be able to:

1. Design and implement digital field surveys for natural resource applications.
2. Demonstrate proficiency in GNSS, TLS, UAV, and ALS workflows for spatial data collection.
3. Process and analyze lidar and imagery to extract ecological and structural information.
4. Integrate and manage spatial datasets using ESRI tools and open-source software.

5. Apply basic machine learning to link field observations with remotely sensed data.
6. Communicate geospatial analyses through dashboards, story maps, and project deliverables.

Required Materials:

There are no required materials for this class. There will be weekly readings posted in the models.

For several of the weekly laboratory sessions, students will be required to use a smartphone or tablet with the Survey123 application (or other designated free applications) installed. Students who do not have access to a suitable device should contact the instructor to arrange for a loaner tablet.

Course Schedule

The schedule below is subject to change.

Module	Focus Of Class
 1	**Intro & Digital Surveys:** • Intro / syllabus / final project intro. • Overview of precision forestry and geospatial tools in natural resource management • Course logistics, safety protocols for fieldwork • Survey123 & Dashboards

2	**GNSS and TLS Intro:** • On Campus collection of GNSS location information along with phone GPS locations • TLS scan intro • Differentially correct data • Visualize point clouds in CloudCompare
3	***Trip to Lubrecht!*** **GNSS & High-Accuracy Data Collection:** • Field data collection (Manual, GNSS, TLS) • GNSS fundamentals: systems, how they work, sources of error, differential correction • Field data collection planning
4	***Trip to Lubrecht!*** **Field Data collection & Drone Intro:** • Field data collection (Manual, GNSS, TLS) • What is photogrammetry? • Drone flight planning and image collection. • UAV regulations, flight planning, overlap requirements.
5	***Trip to Lubrecht!*** **Drone data acquisition and 3D images:** • Field data collection (Drone Imagery) • Drone Photogrammetry and TLS. • Getting metrics from the 3d images. Similarities and differences.

6	***Trip to Lubrecht!*** **ALS Intro:** • Field data collection (Wetland Delineation) • ALS basics and how lidar actually works and it's applications. • Where to get ALS. Sources of ALS (USGS 3DEP, state datasets)
7	***Last trip to Lubrecht!*** **Scale and Data sources:** • Field data collection (Group Determination) • What are high resolution sources of data? • How does scale impact the information you can derive.
8	**Point Cloud Matching:** • Aligning all of our datasets • Projection and scale issues • Using plot data and drone and TLS as ground truth for models using ALS.
9	**R for lidar processing:** • Intro to LidR. • Creation of DTMs, DSMs, CHMs… etc • Metrics derivable from these DEMs
10	**Grid Metrics:** • Rasterization of Lidar data • Percentiles, canopy closure, point cloud normalization

11	**Tree Metrics:** • Tree Approximate Objects (TAOs) • Tree segmentation both for ALS and TLS • Intro to TreeLS and Spanner
12	**Regression and Random Forest:** • Ground truth data related to rasters from lidar • Pairwise regression analysis • Random forest and basic machine learning (ML)
13	**Ground Cover Classification:** • Isolating ground only images from photogrammetry • ML for land cover classification at a fine scale • Accuracy assessments and confusion matrices
14	**Review & Final Project:** • Deeper dive into class determined topic • Class time for final project work

Weekly Expectations

Each Week will consist of:

- Field trips to Lubrecht Experimental Forest.
    - The first part of the semester we will be spending most lab sessions out in the field. Transportation will be provided.
- Lectures on module concepts
- Individual work time to complete Module assignments.

    - Students will be expected to submit individual labs but they can be worked on collectively. They will require out-of-class time to finish (Homework).

A final project will be completed utilizing concepts from the class. More information about the final project is available on the corresponding module page.

Assignments are due at 11:59 PM on the Wednesday after they were assigned.

LATE policy: <u>A 5% deduction in points will be applied PER DAY the assignment is late up to a 50% reduction in grade.</u> Exceptions to this policy may be made with prior arrangement.

Credits & Grading

3 credits will be awarded upon successful completion of this course

A : 93 – 100%, A- : 90 – 92.99%, B+ : 87 – 89.99%, B : 83 – 86.99%, B- : 80 – 82.99%, C+ : 77 – 79.99%, C : 73 – 76.99%, C- : 70 – 72.99%, D+ : 67 - 69.99%, D : 63 – 66.99%, D- : 60 - 62.99%, F : 59.99% and below

Course point breakdown (100 points. 1 point = 1% of final grade):

- 13 weekly assignments worth 5 points each: 78 points possible
    - Some assignments will have 'Extra Credit' worth additional points.
    - Extra credit assignments will be in some of the answer keys you are provided after you submit your work. Not all answer keys will have extra credit options.
- A final project (22 points)
    - 1 point for posting your topic
    - 1 point for posting your data sources
    - 15 points for the project itself based on a provided rubric
    - 5 points for providing peer review of others projects
        - Note that you can't be assigned peer review unless you submit your own final project **on time**.

Graduate Increment

In addition to the work required of all students, graduate students must also complete a short literature review focused on one of the course modules. They will present their review and findings to the class. This component will account for 20% of the final project grade.

There are 100 points for the course. With the extra credit (contained within the answer keys), it is possible to get more than 100 points in this class. Late assignments will make obtaining credit much harder.

Assignments are due at 11:59 PM on the Wednesday after they were assigned.

Homework: You can expect 4 hours of homework per week for this class (much more if you miss the in-person class). You can either work in the campus computer labs or on your own computer if it is powerful enough.

Computer specifications for ArcGIS Pro software.

ArcGIS downloading and installation instructions.

Campus Computer Labs:

Rooms 218 and 219 in Stone Hall are dedicated computer labs supporting coursework and research in geography, conservation, and climate science. Room 219 provides swipe access via Griz Card and connects internally to Room 218, allowing students to move between labs when permitted. Labs may be accessed outside scheduled class times, and Stone Hall is unlocked during regular business hours (8:00 AM–5:00 PM, Monday through Friday).
Students enrolled in this course will automatically receive card access. Students are expected to follow posted lab rules and maintain a respectful academic environment.

Other campus computer lab options are:

- Library main floor computers as well as some of the studio computers.
- Computer lab Native American 014

Responsibilities

We are entering a social contract together as we begin this academic semester. We each have responsibilities to one another. Responsibilities as a teacher and student. Some of these responsibilities are:

Teacher Responsibilities

- **Treat everyone in the room with respect.**
- **Maintain clear, unbiased, and unambiguous expectations for assessment.**
    - **Formative assignments** (e.g., weekly quizzes, in-class activities) are low-stakes opportunities for practice, correction, and discussion. They highlight important concepts and help you prepare for higher-stakes work.
    - **Summative assignments** (e.g., weekly labs, final project) carry the majority of points in the course. Strong performance on these is necessary to demonstrate subject mastery and receive credit.
- **Respond to student communication promptly.**
  Replies will be made within 48 hours (excluding weekends). Additional support will be provided during office hours.
- **Use Canvas announcements** to inform students of any schedule or assignment changes.
- **Foster an inclusive learning environment** that accommodates diverse learners.
  Please share any access needs with me, and feel free to move around or use silent fidgets during class.
- **Be committed to listening to student concerns** about the course, myself, or others, and address them respectfully and constructively.

Student Responsibilities

- **Treat everyone in the room with respect.**
  This includes engaging professionally, valuing diverse perspectives, and listening actively. Bigotry of any kind, or dismissive or demeaning comments about another person's identity (including race, ethnicity, gender, sexual orientation, cultural upbringing, disability, or other personal attributes) will not be tolerated.
- **Take responsibility for your own learning.**

Metacognition (click the link for more info!) —"thinking about thinking"—means being aware of your learning needs and assessing your goals. Use campus resources (linked below) to support your success, and seek help when needed.

- **Stay current with course communications.**

    - Monitor Canvas announcements and messages, and ensure they are forwarded to your email if necessary.
    - Respond to direct communication within 48 hours (excluding weekends).
    - Familiarize yourself with the Canvas course structure.
- **Fully participate** in class activities and assignments.
- **Submit all work on time.**
- **Uphold academic honesty** as defined in the policy below.
- **Read all assignment feedback** carefully and use it to improve your work.
- **Provide constructive feedback** about the course, including what has worked well and what could be improved.

Note: Additional responsibilities may arise during the course. This list is not exhaustive. Please communicate any questions or concerns promptly.

Use of Artificial Intelligence (AI) Tools

AI tools such as ChatGPT, Copilot, or similar platforms are increasingly part of professional workflows in GIS, remote sensing, and cartography. In this course, you may use AI tools as a support for learning, provided that you follow these guidelines:

- **Permitted uses:** You may use AI tools to help brainstorm project ideas, debug R or Python code, clarify GIS concepts, or practice writing.
- **Prohibited uses:** You may not submit AI-generated maps, text, or code as your own without modification or acknowledgment.
- **Attribution:** If you use AI in your work, include a brief note describing how it was used (e.g., “Used ChatGPT to troubleshoot ArcGIS Pro error messages and adapted the suggested Python snippet to my dataset”).
- **Responsibility:** You are responsible for verifying the accuracy of any AI output. These tools often produce incorrect or misleading results, especially with technical topics like coordinate systems, projections, and statistical methods.
- **Instructor discretion:** Certain labs or assignments may explicitly prohibit AI use. Any restrictions will be stated clearly in the assignment instructions.

The purpose of this policy is not to discourage AI use, but to ensure that it supports your learning and that your submitted work reflects your own understanding of GIS concepts and practices.

University Policies

Student Conduct Code and Community Standards

The Student Conduct Code at the University of Montana embodies and promotes honesty, integrity, accountability, rights, and responsibilities associated with constructive citizenship in our academic community. This Code describes expected standards of behavior for all students, including academic conduct and general conduct, and it outlines students' rights, responsibilities, and the campus processes for adjudicating alleged violations.

ODE Accommodations

The University of Montana assures equal access to instruction through collaboration between students with disabilities, instructors, and the Office for Disability Equity (ODE). If you anticipate or experience barriers based on disability, please contact the ODE at 406-243-2243 or ode@umontana.edu, or visit Office of Disability Equity website for more information.

Land Acknowledgement

The University of Montana acknowledges that we are in the aboriginal territories of the Salish and Kalispel people. Today, we honor the path they have always shown us in caring for this place for the generations to come.

Technology & Canvas Support

The University of Montana has designed a course for students to learn the basics of Canvas. UM Passport to Canvas is a self-paced online course to learn everything you need to know about using Canvas in your classes!

Canvas Resources for Students is another resource with curated content that includes videos, guides, and detailed instructions covering information on Assignments Discussions, Quizzes, and Grades.